

The Strengths and Limitations of Quantum Computers

Quantum computing is an emerging industry, presenting opportunities for any fields requiring a high degree of computational power such as AI, material science, and medical research. However, quantum computers suffer from disadvantages which conventional computers do not, making their construction expensive and increasing their error rate.

Basics of a Quantum Computer

The qubit, which corresponds the bit in classical computing, exists in a state of superposition until it is measured. Quantum computers may use the spin state of an electron, or the polarisation of a photon as horizontal or vertical, or even the time taken for a photon to arrive as either early or late – all of which are binary, satisfying our requirement for 2 states per qubit exactly. With reference to the lecture notes¹, the quantum state of a qubit:

$$|\chi\rangle = \sum_n c_n |\chi_n\rangle$$

Where c_n^2 is the probability of observing the state χ_n . Unfortunately, checking on the qubits as they pass the through logic gates, to keep track of calculations, would disrupt the computation as the qubits collapse to a single state – causing the loss of the complex values stored in the probability amplitudes c_1, c_2 (*specification information*) in the state equation above.

Quantum Advantage

As the number of qubits is increased, the number of possible states of the increases exponentially.

$$n = 2^{no. qubits}$$

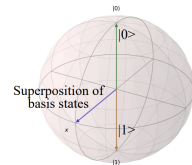
Quantum computers require that qubits are in the *Bell State*, which is the state of maximal entanglement between two qubits².

$$|\Phi^\pm\rangle = \frac{1}{\sqrt{2}}(|00\rangle \pm |11\rangle)$$

$$|\Psi^\pm\rangle = \frac{1}{\sqrt{2}}(|01\rangle \pm |10\rangle)$$

Since the Bell State remains invariant under a *local unitary transformation*, in other words, a quantum logic gate, they can contribute to the computational power. Quantum logic gates can be considered to be a transformation

of our eigenvectors in Hilbert space, where those eigenvectors are the basis vectors.³ Each point on the Bloch sphere below represents a state of the qubit:



Since the qubits are entangled, an operation on a single qubit will **instantaneously** affect other qubits. Therefore, the computational power increases exponentially with the number of qubits. There are, however, challenges to exponentially increasing computational power by scaling the number of qubits.

Challenges to Quantum Supremacy

The Bell State between two qubits can be disrupted if the qubits become entangled with the environment, causing classical-like behaviour in the qubits – this is called **decoherence**. To prevent this effect, quantum computers must be isolated from electromagnetic fields (via a Faraday cage), any vibrations, and they must be supercooled to near absolute zero.⁴

In addition, increasing the number of qubits increases the error rate due to **decoherence**. Similarly, the logic gates themselves can contribute to the noise in quantum computations. In 2019, a group at Penn State University determined that the set up of the gates and the qubits used (the topology) can determine the noise generated.⁵ The group found that optimising the number of gates, and the connectivity of the qubits (low depth topology), could decrease noise by 20

The error rate can also be reduced very effectively on the software side. An Australian group was able to reduce the error rate in quantum logic gates by an order of magnitude.⁶

¹Lecture 6 – Spin. Equation 6.3

²As opposed to weak entanglement, when there is merely a correlation and not a certainty between the states of two qubits.

³Lecture 4 - Matrix Mechanics Equation page 2

⁴<https://news.mit.edu/2023/new-way-quantum-computing-systems-keep-their-cool-0221>

⁵<https://arxiv.org/abs/1904.04323>

⁶<https://www.sydney.edu.au/news-opinion/news/2019/02/28/sydney-physicists-use-code-to-reduce-quantum-error-in-logic-gate.html>